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Chapter 1

Introduction

Part I

Effective Action and Effective Potential

Chapter 2

General Formalism of the Effective Action

The central object of this thesis is the *effective action* $\Gamma[\varphi_c]$, a functional that encodes the full quantum dynamics of a field theory in a single object. Before defining it, we build up to it systematically: starting from the generating functional $Z[J]$ for all Green functions, passing through the connected generating functional $W[J]$, and finally arriving at $\Gamma[\varphi_c]$ via a Legendre transform. The payoff is that the equation of motion derived from Γ governs the quantum-corrected vacuum structure of the theory, making it the natural tool for studying spontaneous symmetry breaking and phase transitions.

2.1 Generating Functionals

Rather than computing Green functions diagram by diagram, all of them, full, connected, and one-particle-irreducible, can be extracted from a small number of functionals by simple functional differentiation.

Coupling the field $\varphi(x)$ linearly to an arbitrary c -number source $J(x)$,

$$\mathcal{L}(\varphi, \partial_\mu \varphi) \longrightarrow \mathcal{L} + J(x) \varphi(x), \quad (2.1)$$

the generating functional for the full Green functions is defined as the vacuum-to-vacuum transition amplitude in the presence of this source,

$$Z[J] = \langle 0 | T \exp \left(i \int d^4x J(x) \varphi(x) \right) | 0 \rangle, \quad (2.2)$$

where $|0\rangle$ is the physical vacuum and $\varphi(x)$ is the Heisenberg-picture field, both defined at $J = 0$. Expanding in powers of J makes the content of this definition explicit,

$$Z[J] = \sum_{n=0}^{\infty} \frac{i^n}{n!} \int d^4x_1 \cdots d^4x_n J(x_1) \cdots J(x_n) G_n(x_1, \dots, x_n), \quad (2.3)$$

where $G_n(x_1, \dots, x_n) \equiv \langle 0 | T[\varphi(x_1) \cdots \varphi(x_n)] | 0 \rangle$ is the full n -point Green function. Each G_n is recovered by n functional derivatives of $Z[J]$ with respect to J , evaluated at $J = 0$. Equivalently, $Z[J]$ admits a path-integral representation,

$$Z[J] = N \int [\mathcal{D}\varphi] \exp \left(i \int d^4x [\mathcal{L}[\varphi(x)] + J(x) \varphi(x)] \right), \quad (2.4)$$

where N is fixed by $Z[0] = 1$. The two representations (2.2) and (2.4) describe the same object and will be used interchangeably throughout.

The full Green functions G_n contain disconnected contributions, products of lower-point functions, that carry no new dynamical information. The *connected* Green functions G_n^c , which isolate only the irreducible correlations, are generated by passing to the logarithm of $Z[J]$,

$$iW[J] = \ln Z[J], \quad (2.5)$$

the proof that this logarithm precisely removes all disconnected pieces is given in Appendix ???. Since $Z[J]$ is the vacuum-to-vacuum amplitude, Eq. (2.5) means

$$e^{iW[J]} = \langle 0^+ | 0^- \rangle_J, \quad (2.6)$$

where $|0^\pm\rangle$ are the vacuum states in the far future and far past, so $W[J]$ is the *vacuum persistence amplitude* in the presence of the external source. This interpretation will be central when we discuss the metastability of the false vacuum in Chapter 7.

The first functional derivative of $W[J]$ defines the *classical field*,

$$\varphi_c(x) \equiv \frac{\delta W}{\delta J(x)} = \left[\frac{\langle 0^+ | \varphi(x) | 0^- \rangle}{\langle 0^+ | 0^- \rangle} \right]_J, \quad (2.7)$$

the source-dependent vacuum expectation value of $\varphi(x)$, which will serve as the argument of the effective action.

2.2 The Effective Action and the Effective Potential

Given $W[J]$ and the classical field φ_c via (2.7), we trade the source J for φ_c as the fundamental variable via the Legendre transform,

$$\Gamma[\varphi_c] = W[J] - \int d^4x J(x) \varphi_c(x), \quad (2.8)$$

where J on the right-hand side is understood as a functional of φ_c obtained by inverting (2.7). Differentiating with respect to φ_c and using (2.7) gives

$$\frac{\delta \Gamma}{\delta \varphi_c(x)} = -J(x). \quad (2.9)$$

In the physical situation $J = 0$, this becomes the quantum equation of motion,

$$\frac{\delta \Gamma}{\delta \varphi_c(x)} = 0, \quad (2.10)$$

the exact analogue of $\delta S / \delta \varphi = 0$ for the classical action, but now with all quantum corrections included. The vacua of the quantum theory are precisely the solutions to (2.10).

Expanding $\Gamma[\varphi_c]$ in powers of φ_c ,

$$\Gamma[\varphi_c] = \sum_{n=0}^{\infty} \frac{1}{n!} \int d^4x_1 \cdots d^4x_n \Gamma^{(n)}(x_1, \dots, x_n) \varphi_c(x_1) \cdots \varphi_c(x_n), \quad (2.11)$$

the coefficients $\Gamma^{(n)}$ are the one-particle-irreducible (1PI) Green functions, or proper vertices, the sum of all Feynman diagrams with n external lines that cannot be disconnected by cutting a single internal propagator, with no external propagators attached.

For a translationally invariant configuration $\varphi_c = \text{const}$, the effective action takes the form $\Gamma = -\Omega V(\varphi_c)$, where Ω is the four-volume. More generally, for slowly-varying fields, Γ admits a derivative expansion,

$$\Gamma[\varphi_c] = \int d^4x \left[-V(\varphi_c) + \frac{1}{2}(\partial_\mu \varphi_c)^2 Z(\varphi_c) + \cdots \right], \quad (2.12)$$

where $V(\varphi_c)$ is the *effective potential*, an ordinary function rather than a functional. Comparing with (2.11), the n -th derivative of V equals the sum of all 1PI diagrams with n external legs at zero external momenta, so the effective potential encodes the full quantum-corrected vacuum structure of the theory. The physical mass and coupling are read off from it via

$$\mu^2 = \left. \frac{d^2 V}{d\varphi_c^2} \right|_{\varphi_c=0}, \quad \lambda = \left. \frac{d^4 V}{d\varphi_c^4} \right|_{\varphi_c=0}. \quad (2.13)$$

2.3 Spontaneous Symmetry Breaking

The effective potential is the natural language for spontaneous symmetry breaking. Suppose $\mathcal{L}$ has a discrete internal symmetry, e.g. $\varphi \rightarrow -\varphi$. This symmetry is spontaneously broken when the quantum equation of motion (2.10) admits a solution with a nonzero, spatially uniform vacuum expectation value,

$$\left. \frac{dV}{d\varphi_c} \right|_{\varphi_c=\langle\varphi\rangle} = 0, \quad \langle\varphi\rangle \neq 0, \quad (2.14)$$

even in the absence of any external source. Stability requires this stationary point to be a minimum of V .

Two phases are possible. In the symmetric phase, $\langle\varphi\rangle = 0$ and the $\varphi \rightarrow -\varphi$ symmetry is intact. In the broken phase, $\langle\varphi\rangle = v \neq 0$ and the symmetry is hidden. The physical spectrum in the broken phase is described by the shifted field $\varphi' = \varphi - v$, and the renormalization conditions (2.13) are displaced to the broken minimum,

$$\mu^2 = \left. \frac{d^2 V}{d\varphi_c^2} \right|_{\varphi_c=v}, \quad \lambda = \left. \frac{d^4 V}{d\varphi_c^4} \right|_{\varphi_c=v}. \quad (2.15)$$

With the formalism in place, we turn in the next chapter to an explicit one-loop computation of the effective potential, first in a simple scalar theory and then in the physically relevant case of scalar electrodynamics.

Chapter 3

Physical Meaning of the Effective Action and Effective Potential

We have defined $\Gamma[\varphi_c]$ and $V_{\text{eff}}(\bar{\varphi})$ formally in the previous chapter. Here we give them concrete physical meaning.

3.1 The Effective Potential as a Generator of 1PI Graphs

For a constant classical field $\bar{\varphi}$, inserting the Fourier representation of $\Gamma^{(n)}$ into the expansion (2.11) gives

$$\Gamma(\bar{\varphi}) = \sum_{n=2}^{\infty} \frac{1}{n!} \bar{\varphi}^n \int \prod_{i=1}^n d^4 x_i \Gamma^{(n)}(x_1, \dots, x_n) \quad (3.1)$$

$$= \sum_{n=2}^{\infty} \frac{1}{n!} \bar{\varphi}^n \int \prod_{i=1}^n \left(d^4 x_i \frac{d^4 k_i}{(2\pi)^4} e^{ik_i x_i} \right) \tilde{\Gamma}^{(n)}(k_1, \dots, k_n) \cdot (2\pi)^4 \delta^{(4)}\left(\sum_{i=1}^n k_i\right), \quad (3.2)$$

where $\int d^4 x e^{ikx} = (2\pi)^4 \delta^{(4)}(k)$ was used. Setting $k = 0$ gives $\int d^4 x = (2\pi)^4 \delta^{(4)}(0)$, so that

$$\Gamma(\bar{\varphi}) = \sum_{n=2}^{\infty} \frac{1}{n!} \bar{\varphi}^n \int d^4 x \tilde{\Gamma}^{(n)}(0, \dots, 0). \quad (3.3)$$

Comparing this with $\Gamma(\bar{\varphi}) = \int d^4 x [-V_{\text{eff}}(\bar{\varphi})]$ gives

$$V_{\text{eff}}(\bar{\varphi}) = - \sum_{n=2}^{\infty} \frac{1}{n!} \bar{\varphi}^n \tilde{\Gamma}^{(n)}(0, \dots, 0). \quad (3.4)$$

Thus $V_{\text{eff}}(\bar{\varphi})$ is the generating functional for 1PI graphs at vanishing external momenta, computed by summing all such graphs and inserting a factor $\bar{\varphi}$ for each external line.

To lowest order in $\hbar$, only tree-level graphs contribute. For a Lagrangian of the form

$$\mathcal{L}(\varphi, \partial_\mu \varphi) = \frac{1}{2} \partial_\mu \varphi \partial^\mu \varphi - U(\varphi), \quad (3.5)$$

each term $g_i \varphi^i / i!$ in $U(\varphi)$ contributes exactly one tree-level 1PI graph, since all tree graphs with more than one vertex are not 1PI. Each term gives a contribution $-g_i \bar{\varphi}^i$, where $-g_i$

is the vertex factor and $\bar{\varphi}^i$ comes from the i external lines. The minus signs cancel and one recovers

$$V_{\text{eff}}(\bar{\varphi}) = U(\bar{\varphi}) + \mathcal{O}(\hbar), \quad (3.6)$$

confirming that the classical potential is the leading-order approximation to the effective potential.

3.2 The Effective Potential as a Minimum Energy Density

A deeper interpretation follows from the variational principle.

Theorem 1. $V_{\text{eff}}(\bar{\varphi})$ is the minimum of the energy density expectation value over all normalized states $|a\rangle$ satisfying $\langle a|\varphi|a\rangle = \bar{\varphi}$. That is,

$$V_{\text{eff}}(\bar{\varphi}) = \langle a|\mathcal{H}|a\rangle, \quad (3.7)$$

for $|a\rangle$ which obeys $\delta\langle a|\mathcal{H}|a\rangle = 0$ subject to the constraints $\langle a|a\rangle = 1$, $\langle a|\varphi|a\rangle = \bar{\varphi}$.

Proof. We first solve the analogous problem in quantum mechanics, then translate to field theory. In QM the problem is to find a state $|a\rangle$ such that $\langle a|H|a\rangle$ is stationary subject to $\langle a|a\rangle = 1$ and $\langle a|A|a\rangle = A_c$ for some self-adjoint operator A . Introducing Lagrange multipliers E and J for each constraint, the problem becomes

$$\delta\langle a|(H - E - JA)|a\rangle = 0 \quad (\text{unconstrained}) \quad (3.8)$$

$$\langle a|a\rangle = 1 \quad (3.9)$$

$$\langle a|A|a\rangle = A_c. \quad (3.10)$$

Since H is self-adjoint, this yields

$$(H - E - JA)|a\rangle = 0, \quad (3.11)$$

with solution $|a\rangle = |a(E, J)\rangle$. Using $\langle a|a\rangle = 1$, E is solved as a function of J , i.e. $E = E(J)$, and $\langle a|A|a\rangle = A_c$ gives $J = J(A_c)$. Thus

$$(H - JA)|a(J)\rangle = E(J)|a(J)\rangle, \quad (3.12)$$

where $|a(J)\rangle$ is a normalized eigenstate of $(H - JA)$. Taking the expectation value with $\langle a(J)|$ and differentiating with respect to J ,

$$\begin{aligned} \left(\frac{\delta}{\delta J} \langle a(J)| \right) \underbrace{(H - E(J) - JA)|a(J)\rangle}_0 + \langle a(J)| \frac{\delta}{\delta J} (H - E(J) - JA)|a(J)\rangle \\ + \langle a(J)| (H - E(J) - JA) \underbrace{\frac{\delta}{\delta J} |a(J)\rangle}_0 = 0, \end{aligned} \quad (3.13)$$

which gives

$$-\frac{\delta E}{\delta J} - \underbrace{\langle a(J)|A|a(J)\rangle}_{A_c} = 0 \quad \Rightarrow \quad \frac{\delta E}{\delta J} = -A_c. \quad (3.14)$$

Substituting back into the expectation value of the eigenvalue equation,

$$\langle a(J)|H|a(J)\rangle = E(J) - J \frac{\delta E}{\delta J}, \quad (3.15)$$

which is the energy of the ground state of H in the space of states satisfying $\langle a(J)|A|a(J)\rangle = A_c$.

We now translate this to field theory. In field theory the only normalizable eigenstate of a Hamiltonian is the vacuum, so $|a(J)\rangle$ is identified with the vacuum of the sourced theory. Adding a source $J(x)\varphi(x)$ to the Hamiltonian density, adiabatically switched on at $t = 0$ and off at $t = T$ over spatial volume V , the adiabatic theorem gives

$$e^{iW(J)} = \langle 0^+ | 0^- \rangle = \exp[-iVT \varepsilon(J)], \quad (3.16)$$

so $-W(J)$ is the ground state action of the sourced theory. The Legendre transform structure,

$$\Gamma(\bar{\varphi}) = W(J) - J \frac{\delta W}{\delta J}, \quad \frac{\delta W}{\delta J} = \bar{\varphi}, \quad (3.17)$$

is formally identical to the quantum-mechanical result above, identifying $-\Gamma(\bar{\varphi})$ as the ground state action of the sourceless theory in the sector with $\langle a|\varphi|a\rangle = \bar{\varphi}$. That is,

$$-\Gamma(\bar{\varphi}) = \langle a|H|a\rangle, \quad (3.18)$$

$$-\frac{\Gamma(\bar{\varphi})}{VT} = \frac{\langle a|H|a\rangle}{VT}, \quad (3.19)$$

and therefore

$$V_{\text{eff}}(\bar{\varphi}) = \langle a|\mathcal{H}|a\rangle \quad (3.20)$$

for the lowest-energy state satisfying $\langle a|a\rangle = 1$ and $\langle a|\varphi|a\rangle = \bar{\varphi}$. $\square$

This result has an important consequence: since $V_{\text{eff}}(\bar{\varphi})$ is defined as a minimum of energy over a convex set of states, it is convex. Its minima therefore correspond to the stable equilibrium configurations of the quantum field theory, a fact that will be central to the analysis of phase transitions in later chapters.

3.3 Variational Definition of the Full Effective Action

The variational interpretation of V_{eff} established above can be generalized to the full effective action.

Theorem 2. $\Gamma(\varphi)$ is the stationary value of

$$\Gamma(\varphi) = \int_{-\infty}^{\infty} dt \langle \psi_-, t | i\partial_t - H | \psi_+, t \rangle \quad (3.21)$$

subject to the constraints

$$\langle \psi_-, t | \Phi(\mathbf{x}) | \psi_+, t \rangle = \varphi(t, \mathbf{x}), \quad \langle \psi_-, t | \psi_+, t \rangle = 1, \quad \lim_{t \rightarrow \pm\infty} |\psi_{\pm}, t\rangle = |0\rangle. \quad (3.22)$$

Proof. The proof follows the same structure as Theorem 2. Introducing Lagrange multipliers $J(t, \mathbf{x})$ and $w(t)$ for the two constraints,

$$\Gamma_m(\varphi) = \int dt \langle \psi_-, t | i\partial_t - H | \psi_+, t \rangle + \int d^4x J(t, \mathbf{x}) (\langle \psi_-, t | \Phi | \psi_+, t \rangle - \varphi(t, \mathbf{x})) - w(t) (\langle \psi_-, t | \psi_+, t \rangle - 1). \quad (3.23)$$

Varying with respect to $\langle \psi_-, t |$, the last term drops out and since $\langle \delta\psi_-, t |$ is arbitrary,

$$\left(i\partial_t - H + \int d^3x J(t, \mathbf{x}) \Phi(\mathbf{x}) \right) | \psi_+, t \rangle = w(t) | \psi_+, t \rangle. \quad (3.24)$$

Varying with respect to $| \psi_+, t \rangle$ and integrating by parts,

$$\int dt \langle \psi_-, t | i\partial_t | \delta\psi_+, t \rangle = [\langle \psi_-, t | \delta\psi_+, t \rangle]_{-\infty}^{\infty} - \int dt i (\partial_t \langle \psi_-, t |) | \delta\psi_+, t \rangle, \quad (3.25)$$

where the boundary term vanishes because $\lim_{t \rightarrow \pm\infty} | \psi_{\pm}, t \rangle = | 0 \rangle$. Hence

$$\int dt \langle \psi_-, t | i\partial_t | \delta\psi_+, t \rangle = - \int dt i (\partial_t \langle \psi_-, t |) | \delta\psi_+, t \rangle. \quad (3.26)$$

Since $| \delta\psi_+, t \rangle$ is arbitrary,

$$\delta_{| \psi_+, t \rangle} \Gamma_m(\varphi) = \int dt \langle \psi_-, t | \left(-i\overleftarrow{\partial}_t + H + \int d^3x J(\mathbf{x}) \Phi(\mathbf{x}) + w(t) \right) | \delta\psi_+, t \rangle = 0, \quad (3.27)$$

and taking the Hermitian conjugate,

$$\left[i\partial_t - H + \int d^3x J(\mathbf{x}) \Phi(\mathbf{x}) \right] | \psi_-, t \rangle = w^*(t) | \psi_-, t \rangle. \quad (3.28)$$

Defining the rescaled states

$$| +, t \rangle = e^{i \int_{-\infty}^t dt' w(t')} | \psi_+, t \rangle, \quad (3.29)$$

$$| -, t \rangle = e^{-i \int_t^{\infty} dt' w^*(t')} | \psi_-, t \rangle, \quad (3.30)$$

these are asymptotic ground states satisfying the sourced Schrödinger equation with no inhomogeneous term. Their overlap gives

$$\langle -, t | +, t \rangle = e^{iW(J)} = \exp \left[i \int_{-\infty}^{\infty} dt w(t) \right], \quad (3.31)$$

from which

$$W(J) = \int_{-\infty}^{\infty} dt \langle \psi_-, t | i\partial_t - H + \int d^3x J \cdot \Phi | \psi_+, t \rangle. \quad (3.32)$$

As shown in Appendix ??, $\delta W(J)/\delta J = \varphi$, so the Legendre transform gives

$$W(J) - \int d^4x J \varphi = \Gamma(\varphi) = \int_{-\infty}^{\infty} dt \langle \psi_-, t | i\partial_t - H | \psi_+, t \rangle, \quad (3.33)$$

which shows that $\Gamma(\varphi)$ is indeed the stationary value. $\square$

Chapter 4

Toy Model: One-Loop Effective Potential and Renormalization

We illustrate the general formalism of the previous chapter with an explicit computation in the simplest setting: a single real scalar field with Lagrangian

$$\mathcal{L}(x) = \frac{1}{2} \partial_\mu \phi(x) \partial^\mu \phi(x) - V(\phi(x)), \quad (4.1)$$

and classical potential

$$V(\phi) = \frac{\alpha_0}{4!} \phi^4 - \frac{m_0^2}{2} \phi^2. \quad (4.2)$$

We work in units $\hbar = c = 1$ throughout.

4.1 The Classical Field and the Effective Action

The classical minimum of $V(\phi)$ sits at some ϕ_0 , but the vacuum expectation value $\langle \phi \rangle$ of the quantum field need not coincide with it. It is defined by

$$\langle \phi \rangle \equiv \lim_{J \rightarrow 0} \frac{\langle 0^+ | \phi_{\text{op}} | 0^- \rangle_J}{\langle 0^+ | 0^- \rangle_J}, \quad (4.3)$$

where the generating functional $W[J]$ encodes the full quantum theory via

$$\exp \frac{i}{\hbar} W[J] = \mathcal{N} \int (\mathcal{D}\phi) \exp \frac{i}{\hbar} \{S[\phi] + (J, \phi)\}, \quad (4.4)$$

with $S[\phi] = \int d^4x \mathcal{L}(x)$ and $(J, \phi) = \int d^4x J(x) \phi(x)$. The classical field, the VEV in the presence of an external source $J(x)$, is

$$\phi_c(x) \equiv \frac{\langle 0^+ | \phi_{\text{op}}(x) | 0^- \rangle_J}{\langle 0^+ | 0^- \rangle_J} = \frac{\delta W[J]}{\delta J(x)}, \quad (4.5)$$

and $\langle \phi \rangle$ is recovered as $\lim_{J \rightarrow 0} \phi_c(x)$. The loop expansion is an expansion in powers of $\hbar c \alpha$, but it performs an infinite resummation of Feynman graphs rather than a perturbative truncation.

The natural question is: given a desired profile ϕ_c , what source J produces it? To answer this, we trade J for ϕ_c as the independent variable via the Legendre transform,

$$\Gamma[\phi_c] \equiv W[J] - (J, \phi_c), \quad (4.6)$$

where J is eliminated in favour of ϕ_c by inverting (3.5). The functional $\Gamma[\phi_c]$ is the effective action, so named because it is a functional of the classical field ϕ_c and thus plays the same role for the quantum theory as $S[\phi]$ does classically. Differentiating (3.6) and using (3.5),

$$\begin{aligned}\frac{\delta\Gamma[\phi_c]}{\delta\phi_c(x)} &= \frac{\delta W[J]}{\delta\phi_c(x)} - J(x) - \int d^4y \frac{\delta J(y)}{\delta\phi_c(x)} \phi_c(y), \\ \frac{\delta W[J]}{\delta\phi_c(x)} &= \int d^4y \frac{\delta W[J]}{\delta J(y)} \frac{\delta J(y)}{\delta\phi_c(x)} = \int d^4y \frac{\delta J(y)}{\delta\phi_c(x)} \phi_c(y),\end{aligned}\quad (4.7)$$

so that

$$\frac{\delta\Gamma[\phi_c]}{\delta\phi_c(x)} = -J(x). \quad (4.8)$$

At $J = 0$, translational invariance forces ϕ_c to be a constant, so $\langle\phi\rangle$ is the root of

$$\left. \frac{d\Gamma[\phi_c]}{d\phi_c} \right|_{\phi_c=\langle\phi\rangle} = 0. \quad (4.9)$$

4.2 The Effective Action as a Generator of 1PI Green's Functions

Expanding $\Gamma[\phi_c]$ in powers of ϕ_c ,

$$\Gamma[\phi_c] = \sum_{n=0}^{\infty} \frac{1}{n!} \int d^4x_1 \cdots d^4x_n \Gamma_n(x_1, \dots, x_n) \phi_c(x_1) \cdots \phi_c(x_n), \quad (4.10)$$

we claim that Γ_n is the n -point 1PI Green function, the sum of all connected 1PI Feynman graphs with n external legs. To see this, consider the functional integral

$$\mathcal{N} \int (\mathcal{D}\phi) \exp \frac{i}{\hbar} \{ \Gamma[\phi] + (J, \phi) \}. \quad (4.11)$$

In the limit $\hbar \rightarrow 0$ the integrand is dominated by its saddle point, which occurs at $\phi = \phi_c$ since (3.8) is precisely the saddle-point condition. Thus

$$\mathcal{N} \int (\mathcal{D}\phi) \exp \frac{i}{\hbar} \{ \Gamma[\phi] + (J, \phi) \} \xrightarrow{\hbar \rightarrow 0} \mathcal{N} \exp \frac{i}{\hbar} \{ \Gamma[\phi_c] + (J, \phi_c) \}, \quad (4.12)$$

and by (3.6) the right-hand side equals $\exp \frac{i}{\hbar} W[J]$, so

$$\exp \frac{i}{\hbar} W[J] \xrightarrow{\hbar \rightarrow 0} \mathcal{N} \int (\mathcal{D}\phi) \exp \frac{i}{\hbar} \{ \Gamma[\phi] + (J, \phi) \}. \quad (4.13)$$

In this limit the right-hand side is the sum of all tree graphs of a theory whose action is $\Gamma[\phi]$, with vertices given by the Γ_n of (3.10). Since $W[J]$ generates all connected Green functions of the original theory, and any connected graph decomposes into 1PI components, it follows that Γ_n is a 1PI Green function of the original theory. $\square$

4.3 The Effective Potential

Alternatively, $\Gamma[\phi_c]$ may be expanded in derivatives of ϕ_c ,

$$\Gamma[\phi_c] = \int d^4x \left[-U(\phi_c(x)) + \frac{1}{2} \partial_\mu \phi_c(x) \partial^\mu \phi_c(x) Z(\phi_c(x)) + \dots \right], \quad (4.14)$$

where $U(\phi_c)$ is the effective potential. For a constant configuration $\phi_c(x) = \rho$ this reduces to

$$\Gamma[\rho] = -\Omega U(\rho), \quad (4.15)$$

where Ω is the total spacetime volume. To connect $U(\rho)$ to the 1PI Green functions, we pass to momentum space. The Fourier transforms of ϕ_c and Γ_n are defined by

$$\tilde{\phi}_c(k) = \int d^4x e^{ik \cdot x} \phi_c(x), \quad (4.16)$$

$$\phi_c(x) = \int \frac{d^4k}{(2\pi)^4} e^{-ik \cdot x} \tilde{\phi}_c(k), \quad (4.17)$$

$$\tilde{\Gamma}_n(k_1, \dots, k_n) (2\pi)^4 \delta^4(k_1 + \dots + k_n) = \int d^4x_1 \dots d^4x_n \Gamma_n(x_1, \dots, x_n) e^{i(k_1 x_1 + \dots + k_n x_n)}. \quad (4.18)$$

Substituting these into (3.10),

$$\begin{aligned} \Gamma[\phi_c] &= \sum_{n=0}^{\infty} \frac{1}{n!} \int d^4x_1 \dots d^4x_n \Gamma_n(x_1, \dots, x_n) \phi_c(x_1) \dots \phi_c(x_n) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \int \frac{d^4k_1}{(2\pi)^4} \dots \frac{d^4k_n}{(2\pi)^4} \int d^4x_1 \dots d^4x_n \Gamma_n(x_1, \dots, x_n) e^{-i(k_1 x_1 + \dots + k_n x_n)} \tilde{\phi}(k_1) \dots \tilde{\phi}(k_n) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \int \frac{d^4k_1}{(2\pi)^4} \dots \frac{d^4k_n}{(2\pi)^4} (2\pi)^4 \delta^4(k_1 + \dots + k_n) \tilde{\Gamma}_n(k_1, \dots, k_n) \tilde{\phi}(k_1) \dots \tilde{\phi}(k_n). \end{aligned} \quad (4.19)$$

Setting $\phi_c(x) = \rho$ gives $\tilde{\phi}_c(k) = (2\pi)^4 \delta^4(k) \rho$, so each momentum integral collapses,

$$\begin{aligned} \Gamma[\rho] &= \sum_{n=0}^{\infty} \frac{1}{n!} \int \frac{d^4k_1}{(2\pi)^4} \dots \frac{d^4k_n}{(2\pi)^4} (2\pi)^4 \delta^4(k_1 + \dots + k_n) \tilde{\Gamma}_n(k_1, \dots, k_n) \\ &\quad \cdot (2\pi)^4 \delta^4(k_1) \rho \dots (2\pi)^4 \delta^4(k_n) \rho \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \underbrace{(2\pi)^4 \delta^4(0)}_{\Omega} \rho^n \tilde{\Gamma}_n(0, \dots, 0), \end{aligned} \quad (4.20)$$

and therefore

$$\Gamma[\rho] = \Omega \sum_{n=0}^{\infty} \frac{\rho^n}{n!} \tilde{\Gamma}_n(0), \quad (4.21)$$

where $\tilde{\Gamma}_n(0) \equiv \tilde{\Gamma}_n(0, \dots, 0)$ and $\Omega = (2\pi)^4 \delta^4(0)$ is the total spacetime volume. Comparing (3.21) with $\Gamma[\rho] = -\Omega U(\rho)$,

$$U(\rho) = - \sum_{n=0}^{\infty} \frac{\rho^n}{n!} \tilde{\Gamma}_n(0), \quad (4.22)$$

which is simply the statement that the Taylor coefficients of U are the zero-momentum 1PI Green functions,

$$-\frac{d^n U(\rho)}{d\rho^n} \Big|_{\rho=0} = \tilde{\Gamma}_n(0). \quad (4.23)$$

The effective potential is thus entirely determined by the 1PI Green functions of the original theory evaluated at vanishing external momenta. In the next section we compute it explicitly at one loop.

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